A SECOND DESCRIPTION OF MODULES OVER A FIBRE PRODUCT OF RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Situation of a fibre product of rings and the functor from modules over the fibre product to pairs of modules. situation at source lines 521-542 and the discussion of the same section, before the counted statement. Supplies the functor whose behaviour is described.
- Base change of modules along the two structure maps and the comparison map between them. used at proof lines 728-740. Provides the two module categories between which the equivalence is asserted.
- Elementwise verification that two functors with a comparison map are inverse to each other. used at proof lines 740-747. Establishes the equivalence of the two descriptions.

Source reference. Original lemma 08IG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10814–10814. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In the situation of Lemma 07RU for a B'-module L' the adjunction map*

$$L' \longrightarrow (L' \otimes_{B'} B) \times_{(L' \otimes_{B'} A)} (L' \otimes_{B'} A')$$

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is surjective but in general not injective.

Proof. As in the proof of Lemma 07RU let $J \subset B'$ be the kernel of the map $B' \to B$. Then $L' \otimes_{B'} B = L'/JL'$. Hence to prove surjectivity it suffices to show that elements of the form $(0, z)$ of the fibre product are in the image of the map of the lemma. The kernel of the map $L' \otimes_{B'} A' \to L' \otimes_{B'} A$ is the image of $L' \otimes_{B'} I \to L' \otimes_{B'} A'$. Since the map $J \to I$ induced by $B' \to A'$ is an isomorphism the composition

$$L' \otimes_{B'} J \to L' \to (L' \otimes_{B'} B) \times_{(L' \otimes_{B'} A)} (L' \otimes_{B'} A')$$

induces a surjection of $L' \otimes_{B'} J$ onto the set of elements of the form $(0, z)$. To see the map is not injective in general we present a simple example. Namely, take a field k, set $B' = k[x, y]/(xy)$, $A' = B'/(x)$, $B = B'/(y)$, $A = B'/(x, y)$ and $L' = B'/(x - y)$. In that case the class of x in L' is nonzero but is mapped to zero under the displayed arrow. $\square$